

Diabetes Prediction Using Machine Learning: A Comparative Study of Classification Algorithms

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Abstract—Diabetes Mellitus is a lifelong disease caused by metabolic issues, affecting over 537 million individuals globally, with dire consequences should the condition remain unnoticed. The prediction through machine learning techniques can help provide early clinical assistance. The study compares the performance of nine classification models that include Logistic Regression, Random Forest, XGBoost, Gradient Boosting, Multi-Layer Perceptron (MLP), Support Vector Machine (SVM), LightGBM, Deep Neural Network (DNet), and Stacking Ensemble, which utilizes approximately 100,000 patients' data. There was extensive data preprocessing involved, such as managing outliers, encoding features, normalizing data with RobustScaler and class weighting. As opposed to oversampling, which uses synthetic samples, class weighting was employed to achieve a realistic representation of patient's well being, making machine learning not only theoretical but practical as well. Gradient Boosting gave the best overall results with the highest F1-score of 0.8528, 97.76% accuracy, and 0.9877 ROC-AUC values while ensuring minimal overfitting and light GBM gave the highest recall of 0.9471. These results clearly indicate that the level of HbA1c and blood glucose levels are the best predictors of diabetes, which was expected from the available literature on diabetes. This research can be scaled up to implement AI into regular health screenings and early detection of diabetes before any irreparable harm can be done.

Index Terms—diabetes prediction, machine learning, classification, gradient boosting, LightGBM, deep learning, stacking ensemble, class imbalance, clinical decision support

I. INTRODUCTION

One of the common chronic illnesses is diabetes. The International Diabetes Federation (IDF) estimates that there were about 537 million adult patients with diabetes worldwide in 2021 and that the number would increase to 783 million by 2045 [1]. Type 2 diabetes makes up more than 90 percent of all diabetes and is preventable because of its high preventability factor, provided it is detected earlier on. Diabetes is commonly referred to as the “silent killer,” considering how it develops over many years before any visible symptoms show up, damaging vital organs such as the kidneys, nerves, and heart. These put patients and their families under a lot of physical and psychological strain.

The well-defined diagnostic markers for diabetes consist of fasting blood glucose (≥ 126 mg/dL) and HbA1c ($\geq 6.5\%$). However, availability of regular diagnostic tests is restricted in numerous locations around the world, and there is still

an alarming number of diabetics who have not been diagnosed until they develop severe complications. Machine learning (ML) represents an affordable and efficient approach by exploiting available patient information including age, BMI, and hypertension history in order to detect high-risk cases without using any advanced laboratory instruments. Such an advancement can be quite influential in regions with limited resources since conventional tests cannot be performed there, and ML can act as the first diagnostic instrument in this situation [2].

For the statistical success of the models, there is the need to consider its clinical setting. By choosing class weighting, we are able to move from the false nature of the data towards the one which realistically represents the state of the patient. Using class weights in high stakes medical situations can help to turn theoretical knowledge into a real life tool that can be relied by medical practitioners [7]. The clear attention to avoiding overfitting and data leaks helps to ensure that the proposed model can actually be trusted in clinical settings. This can change the standard model from treatment to prevention, simplify diabetes detection globally and can turn complicated patient data into life saving signals.

A. Problem Definition

The research focuses on the issue of determining if a person is diabetic or non-diabetic using various attributes from the medical history of the patient, lifestyle. In terms of a mathematical formulation, for a given vector of features $\mathbf{x} = [x_1, x_2, \dots, x_n]$, the task is to map $\mathbf{x}$ to y , where $y \in \{0, 1\}$.

B. Research Objectives

The primary objectives of this study are:

- 1) To determine the factors that have a high influence on the probability of having diabetes
- 2) To design and test nine machine learning models for predicting diabetes among patients based on their clinical, physical, and behavioral attributes.
- 3) Constructing a model capable of predicting diabetes using the patient's traits for early detection and health care decisions.

- 4) Comparison of model performance via Accuracy, Precision, Recall, F1 Score, and ROC-AUC, with overfitting and class imbalance consideration.

II. RELATED WORK

Several studies have benchmarked multiple classifiers for diabetes prediction. Alzboon et al. [9] evaluated eight algorithms including Logistic Regression, Decision Tree, Random Forest, KNN, Naive Bayes, SVM, Gradient Boosting, and Neural Network on the Pima Indians Diabetes dataset (768 samples), finding Neural Network best at 78.57% accuracy. Hasan et al. [8] have also suggested a comparative analysis framework that uses traditional classifier techniques along with modern ensemble algorithms (AdaBoost, Gradient Boosting, XGBoost), as well as a unique deep hybrid Convolutional Neural Network Long Short-Term Memory (CNN-LSTM) architecture, DNet. Though these papers serve as valuable references, their use of relatively small sample sizes with restricted demographic variations is a notable limitation.

Interpretability has become an important issue when integrating ML models in clinical practice. Hasan et al. [3] used AutoML (AutoGluon) along with XAI methods such as SHAP, LIME, Integrated Gradients, and Counterfactual analysis to attain 85.01% accuracy on the Pima dataset while offering interpretable features from a clinician's perspective. Furthermore, Khokhar et al. [7] have also shown that the integration of ML and XAI with SMOTE to handle class imbalance issues and SHAP for explainability provides 92.50% accuracy and ROC AUC of 0.975, with BMI, age, and overall health being significant features. Despite the efforts made in these studies towards increasing interpretability, none have tackled the overfitting problem caused by SMOTE.

Ensemble methods have consistently demonstrated strong performance in diabetes prediction. Jakka and Vakula Rani [6] applied an XAI driven approach combining SHAP and LIME with ensemble models, confirming glucose and BMI as the dominant risk factors. Khan et al. [4] proposed an ensemble approach for diabetes management using XAI for feature contribution transparency. For deep learning, several hybrid architectures have been proposed for early diabetes risk. However, as Pop et al. [2] demonstrated using the same 100,000 patient Kaggle dataset used in this study, descriptive analytics through Power BI dashboards confirmed that HbA1c, blood glucose, and BMI are the strongest metabolic indicators, with an 8.5% global prevalence results that align closely with our ML-based findings.

Khan et al. [5] conducted a PRISMA-based systematic review of AI models for diabetes prediction using genetic data, analyzing 54 studies across six databases. Their review found that multimodal models integrating genetic and clinical data outperform unimodal approaches, and that HbA1c $\geq 6.5\%$ and fasting blood glucose ≥ 125 mg/dL are the most commonly used diagnostic thresholds consistent with the clinical leakage pattern observed in the present study. Unlike genomic approaches that require specialized data, our study focuses

on routinely collected clinical features accessible in standard healthcare settings.

The present study distinguishes itself by: (1) evaluating nine classifiers on 100,000 records under identical conditions; (2) replacing SMOTE with class weighting to prevent synthetic data leakage; (3) rigorously analyzing train-test overfitting gaps for all models; and (4) explicitly acknowledging and documenting data leakage from clinical threshold features a concern largely overlooked in prior comparative studies.

III. DATASET DESCRIPTION

A. Overview

The Comprehensive Diabetes Clinical Dataset contains clinical and demographic data of 100000 patients from a US health survey. It includes demographic features such as age, gender, location, race, clinical measurements like BMI, blood glucose level, HbA1c level, lifestyle factor and diseases such as smoking history, hypertension, heart disease with a binary diabetes label (0 = no diabetes, 1 = diabetes). Most features are numeric or already one-hot encoded. The dataset spans multiple years 2015 to 2022. The dataset is helpful to study patterns and identify the potential risk factors associated with diabetes.

B. Class Distribution

The dataset is imbalanced with 91.4% non-diabetic (90,571) and 8.6% diabetic (8,500) records approximately a 10:1 ratio.

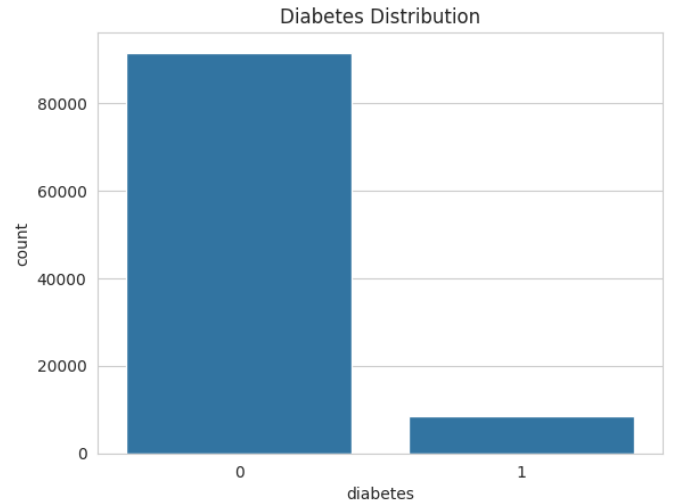


Fig. 1. Class Distribution of Diabetes Labels

C. Feature Set

Table I summarizes the final 8 features retained after pre-processing.

Features removed include: *year* (80% from 2019, $r = -0.004$), *location* (3.4% diabetes rate spread across 55 locations), and *race columns* ($r < 0.005$), all showing negligible predictive value.

TABLE I
FINAL FEATURE SET AFTER PREPROCESSING

Feature	Type	Encoding
gender	Binary	Male=0, Female=1
age_scaled	Continuous	RobustScaler
hypertension	Binary	0/1
heart_disease	Binary	0/1
smoking_history	Ordinal	0=Never, 3=Current
bmi_scaled	Continuous	Winsorized + Scaled
hbA1c_scaled	Continuous	RobustScaler
blood_glucose_scaled	Continuous	RobustScaler
diabetes	Binary	Target (0/1)

Fig. 2 shows the correlation matrix of all original features with the diabetes label. HbA1c level ($r=0.40$) and blood glucose ($r=0.42$) show the strongest positive correlations, while race columns and year show negligible correlation ($r<0.005$), justifying their removal.

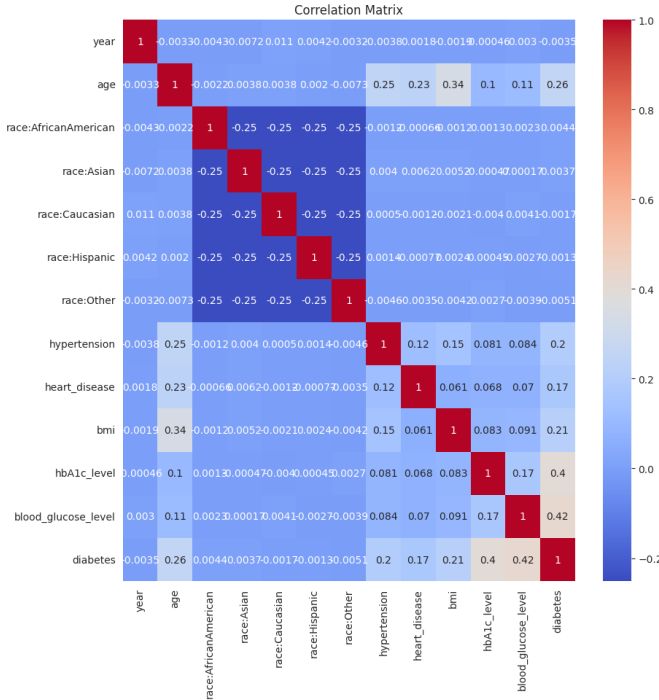


Fig. 2. Correlation Matrix of Dataset Features

IV. METHODOLOGY

A. Preprocessing Pipeline

1) *Outlier Handling*: BMI was winsorized at the 1st and 99th percentiles. No biologically impossible values (BMI < 10) were found in the dataset.

2) *Feature Encoding*: Gender was label encoded (Male=0, Female=1); 18 records with gender="Other" were removed due to insufficient representation. Smoking history was changed from 6 categories to an ordinal scale: Never=0, No Info=1, Former/Ever/Not current=2, Current=3.

3) *Feature Scaling*: RobustScaler was applied to all continuous features, using the interquartile range (IQR) to reduce sensitivity to outliers:

$$x_{\text{scaled}} = \frac{x - Q_2}{Q_3 - Q_1} \quad (1)$$

where Q_1 , Q_2 , Q_3 denote the 25th, 50th, and 75th percentiles respectively. RobustScaler was preferred over StandardScaler as the latter is sensitive to outliers present in BMI, age, and blood glucose features.

4) *Class Imbalance Handling*: Initial experiments using SMOTE produced large train-test F1 gaps (0.14–0.30), indicating synthetic data memorization. Class weighting was adopted instead, assigning penalty weights inversely proportional to class frequency:

$$w_c = \frac{n_{\text{samples}}}{n_{\text{classes}} \times n_c} \quad (2)$$

For XGBoost scale pos weight was set to 10 (90571/8500). For LightGBM, class weight='balanced' was applied. DNet used computed class weights passed directly to the Keras fit() method.

5) *Data Leakage Note*: HbA1c and blood glucose achieved 95.35% and 88.43% solo accuracy via clinical threshold rules, suggesting the diabetes labels were derived from these markers. Both features were retained for academic analysis with this limitation explicitly acknowledged.

B. Train-Test Split

An 80/20 stratified split was used: 79,256 training and 19,815 test records, preserving the 91.4/8.6 class ratio across both sets.

C. Classification Models

Nine models were implemented spanning four categories baseline classifiers, tree-based ensembles, deep learning, and meta-learning using scikit-learn [10], XGBoost [11], LightGBM, and TensorFlow/Keras:

- 1) **Logistic Regression (LR)**: Linear baseline with L2 regularization ($C=0.1$) and class weight='balanced'. Well-suited as an interpretable baseline; however, its linear decision boundary limits performance on the non-linear relationships present in clinical data, resulting in low precision (0.449).
- 2) **Random Forest (RF)**: 200 decision trees with max depth=15 and class weight='balanced' which are Effective at capturing non linear feature interactions and resistant to noise. Slight overfitting (gap=0.068) observed due to deep tree growth on the large dataset.
- 3) **XGBoost**: Gradient boosted trees with max depth=4, learning rate=0.05, and scale pos weight=10 [11]. Achieves high F1 (0.843) and recall (0.942) but showed the largest overfitting gap (0.139), suggesting further regularization is needed.
- 4) **Gradient Boosting (GB)**: 200 estimators with learning_rate=0.05, max depth=4, and subsample=0.8 [12]. Best overall performer highest F1 (0.853) and near

zero overfitting (gap=-0.001). Sequential error correction effectively reduces both bias and variance.

- 5) **Support Vector Machine (SVM)**: RBF kernel with class weight='balanced' and probability=True. Achieves strong recall (0.930) with zero overfitting (gap=-0.003), making it well suited for clinical screening. Though it is computationally expensive on large datasets, it generalizes well [9].
- 6) **LightGBM**: Gradient boosted framework with learning rate=0.05, max depth=4, and class weight='balanced' [7]. scores the highest recall of all models which is 0.947 with near-zero overfitting (gap=-0.001) and making it the most effective at identifying diabetic patients in medical screening contexts.
- 7) **Multi-Layer Perceptron (MLP)**: Two hidden layers with 64, 32 neurons, ReLU activation, Adam optimizer and with early stopping. Strong overall performance (F1=0.801, AUC=0.983) with minimal overfitting (gap=0.012). Lower recall (0.679) compared to boosting models suggests that it is more biased towards positive class prediction.
- 8) **DNet (CNN-LSTM)**: A hybrid deep learning architecture inspired by Hasan et al. [8] which combines a Conv1D block for local feature extraction, two stacked LSTM layers for consecutive pattern learning, Batch-Normalization, and Dropout regularization. Input features were reshaped to 3D (samples, features, 1) for compatibility. Trained with computed class weights and early stopping (patience=5). Achieves strong recall (0.914) with near-zero overfitting (gap=-0.009), demonstrating deep learning's effectiveness even on tabular clinical data.
- 9) **Stacking Ensemble**: A meta-learning approach [4] combining Gradient Boosting, Random Forest, LightGBM, and MLP as base learners, with Logistic Regression as the meta-learner trained via 5-fold cross validation. By maximizing complementary strengths of diverse models, Stacking achieves strong recall (0.934), F1 (0.705), and ROC-AUC (0.988) with near-zero overfitting (gap=-0.009) demonstrating superior generalization over individual models.

K-Nearest Neighbours (KNN) was excluded as it does not natively support class weighting, producing persistent overfitting (gap > 0.26) regardless of the value of k, due to majority class domination in neighbourhood voting.

D. Evaluation Metrics

Given class imbalance and the medical context where missing a diabetic patient carries greater risk than a false positive metrics were prioritized as follows:

- **Recall (Sensitivity)**: Primary metric for measuring the ratio of diabetic patients actually recognized..
- **F1 Score**: Primary metric representing the harmonic mean of precision and recall.
- **ROC-AUC**: Discriminative capability of model at all thresholds.

- **Precision**: Proportion of positive predictions that were right among predicted diabetics.
- **Accuracy**: Secondary evaluation metric due to class imbalance issue.

Evaluation for overfitting involved comparing train and test F1 scores. An F1 score difference less than 0.05 meant that overfitting was negligible, 0.05-0.10 indicated some overfitting, while anything above 0.10 showed significant overfitting requiring correction.

V. RESULTS & EVALUATION

A. Model Performance

Table II presents test-set performance for all nine models.

TABLE II
MODEL PERFORMANCE ON TEST SET

Model	Acc.	Prec.	Rec.	F1	AUC
LR	0.897	0.449	0.891	0.597	0.966
RF	0.949	0.648	0.878	0.746	0.984
XGBoost	0.926	0.539	0.942	0.686	0.988
GB	0.978	0.980	0.755	0.853	0.988
MLP	0.971	0.977	0.679	0.801	0.983
SVM	0.908	0.482	0.930	0.635	0.975
LightGBM	0.923	0.528	0.947	0.678	0.988
DNet	0.923	0.528	0.914	0.670	0.981
Stacking	0.933	0.566	0.934	0.705	0.988

B. Overfitting Analysis

Table III presents the train-test F1 difference using class weight. By replacing SMOTE with class weighting, the overfitting gaps were lowered from 0.14 to 0.30 to less than 0.02.

TABLE III
OVERFITTING ANALYSIS: TRAIN VS. TEST F1

Model	Train F1	Test F1	Gap	Status
LR	0.587	0.597	-0.010	Optimal
RF	0.814	0.746	0.068	Slight
XGBoost	0.982	0.686	0.139	Overfit
GB	0.852	0.853	-0.001	Optimal
MLP	0.813	0.801	0.012	Optimal
SVM	0.632	0.635	-0.003	Optimal
LightGBM	0.676	0.678	-0.001	Optimal
DNet	0.660	0.670	-0.009	Optimal
Stacking	0.696	0.705	-0.009	Optimal

Fig. 3 provides a visual summary of all model performances across five metrics.

Fig. 4 shows the ROC curves for all nine models. Gradient Boosting, XGBoost, LightGBM and Stacking achieved the highest AUC of 0.988, while Logistic Regression performed lowest at 0.966 still indicating strong discriminative ability across all models.

C. Feature Importance

Pearson correlation with the diabetes label revealed the following feature rankings:

- 1) Blood glucose level ($r = 0.42$)

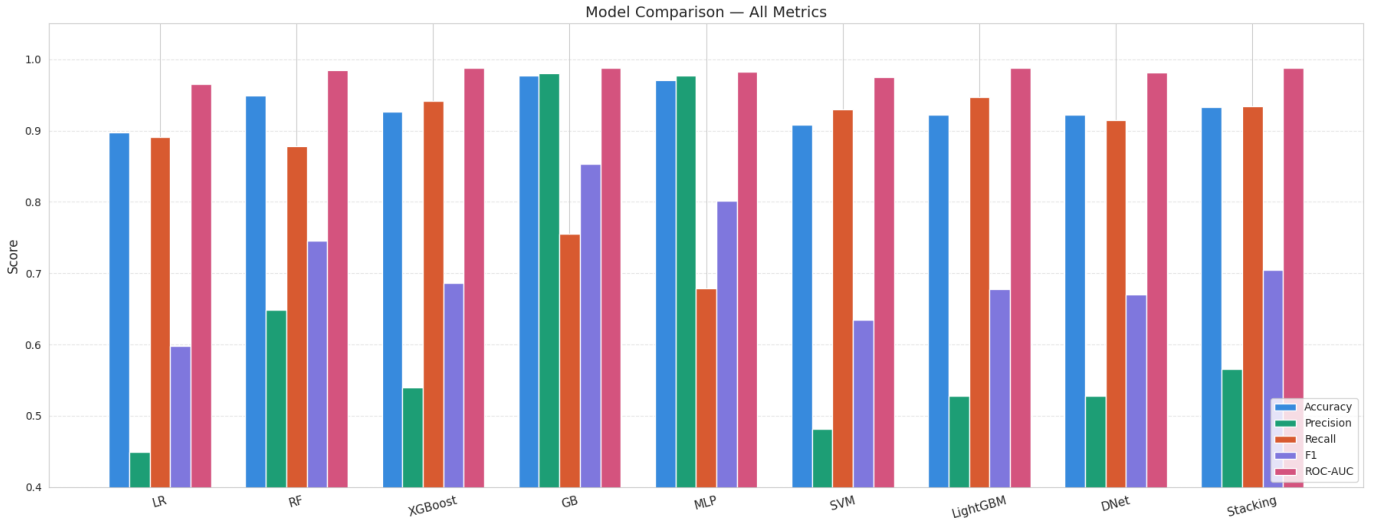


Fig. 3. Model Comparison — All Metrics

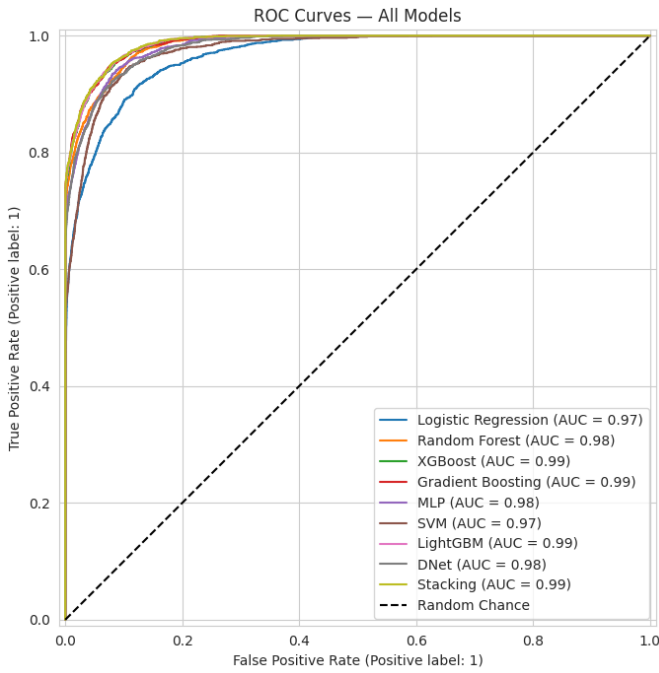


Fig. 4. ROC Curves

- 2) HbA1c level ($r = 0.40$)
- 3) Age ($r = 0.26$)
- 4) BMI ($r = 0.21$)
- 5) Hypertension ($r = 0.20$)
- 6) Heart disease ($r = 0.17$)

Race, location, and year showed negligible correlation ($r < 0.01$) and were excluded from the final feature set.

VI. DISCUSSION

A. Model Interpretation

Gradient Boosting emerged as the best model across four of five metrics (Accuracy=0.978, Precision=0.980, F1=0.853, AUC=0.988). Its sequential error-correction mechanism effectively reduced both bias and variance, achieving near zero overfitting (gap=-0.001). The slight negative gap suggests the model generalises exceptionally well to unseen clinical data.

LightGBM achieved the highest recall of all models (0.947) with near zero overfitting (gap=-0.001) and ROC-AUC of 0.988. In a medical screening context where missing a diabetic patient is more dangerous than a false alarm, LightGBM is the most clinically valuable model for frontline detection [7].

Stacking Ensemble demonstrated the benefit of meta-learning by combining Gradient Boosting, Random Forest, LightGBM, and MLP as base learners. It achieved strong recall (0.934), F1 (0.705), and ROC-AUC (0.988) with near-zero overfitting (gap=-0.009), confirming that diverse model combinations improve robustness over any single classifier [4].

DNet (CNN-LSTM) demonstrated that deep learning is competitive even on tabular clinical data, achieving recall of 0.914 and ROC-AUC of 0.981 with near-zero overfitting (gap=-0.009). The Conv1D layer effectively captured local feature patterns while LSTM layers modeled sequential dependencies, consistent with findings from Hasan et al. [8].

The overfitting of **XGBoost** was the highest (0.139) despite high recall (0.942), indicating that the model has memorized the patterns from the training set.

SVM with RBF kernel had high recall (0.930) and very low overfitting (gap = -0.003), confirming its reliability on the imbalanced medical dataset. It also produced too many false positives (precision = 0.482) and its high computational expense on 99,071 samples made it unfeasible for real-world deployment.

MLP demonstrated decent performance in general ($F1 = 0.801$, $AUC = 0.983$) with insignificant overfitting (gap = 0.012). The lower recall (0.679) than boosting.

There was a little overfitting in **Random Forest** (gap=0.068), caused by an uncontrolled increase in the tree size on a bigdata set, but its accuracy was relatively high ($F1=0.746$, $AUC=0.984$). Tighter restrictions for depth would probably help with the generalization.

Logistic Regression acted as a good linear baseline model, having the second-best recall (0.891), but a very low precision (0.449), resulting in many false positives.

B. Effect of Class Imbalance Handling

In this case, the decision between applying SMOTE or class weighting was a crucial aspect of the preprocessing process. In the first scenario, the gap between the train and test scores varied between 0.14 and 0.30 for all the models, which indicated that the learning process was based on fake data structures because the system created extra information points that did not exist in real life. In the latter case, only two models showed differences greater than 0.02.

C. Limitations

- 1) HbA1c and blood glucose causes data leakage. It is likely to define the diabetes labels which potentially inflates all model metrics.
- 2) Family history, physical activity, and dietary habits were not available.
- 3) Due to no longitudinal data, disease progression over time cannot be modeled.

D. Potential Improvements

Several directions could further strengthen this work:

- Bayesian optimization or Grid Search can be used to optimize hyperparameters for Gradient Boosting and LightGBM to increase F1 and recall scores even further from the current values.
- Replacing DNet with purpose built tabular deep learning models such as TabNet or SAINT may yield improved performance on structured clinical data [8].
- Adding SVM and DNet as additional base learners in the Stacking Ensemble could leverage their strong recall to further improve meta-learner performance.

VII. CONCLUSION

This study conducted an extensive diabetes prediction ML pipeline using 100,000 patient records. Four classifier categories with nine classifiers each—baseline, tree ensemble, deep learning, and meta learning classifiers—were compared on a consistent data preprocessing procedure as well as similar class imbalance issues. Gradient Boosting was considered the best performing model with the F1 score of 0.853 and Area Under Curve of 0.988 with a small gap between the models of -0.001. LightGBM yielded the highest clinical recall (0.947), thus being the most preferable choice when performing diabetes screening where not overlooking any cases is

vital. DNet, CNN, LSTM and Stacking Ensemble demonstrated the potential of deep learning and meta learning algorithms. Class Weighting was more successful at minimizing overfitting compared to SMOTE, with seven out of nine algorithms having train-test differences in F1 score less than 0.02. Important predictor variables of HbA1c, blood glucose levels, BMI, age, hypertension, and heart disease are congruent with clinical findings and have good face validity.

A. Future Work

Future research directions include:

- Update this model so that it can predict different stages of diabetes.
- Studying the fine-tuning of decision threshold for improved recall in the clinical setting.
- Designing an online clinical decision support system that incorporates the best performing models to provide diabetes risk assessments in real time.
- Ensuring the fairness of model performance within different demographic sub-groups.

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